PIC 10A 1A

TA: Bumsu Kim

This content is protected and may not be shared, uploaded, or distributed.

The author does not grant permission for these notes to be posted anywhere without prior consent.

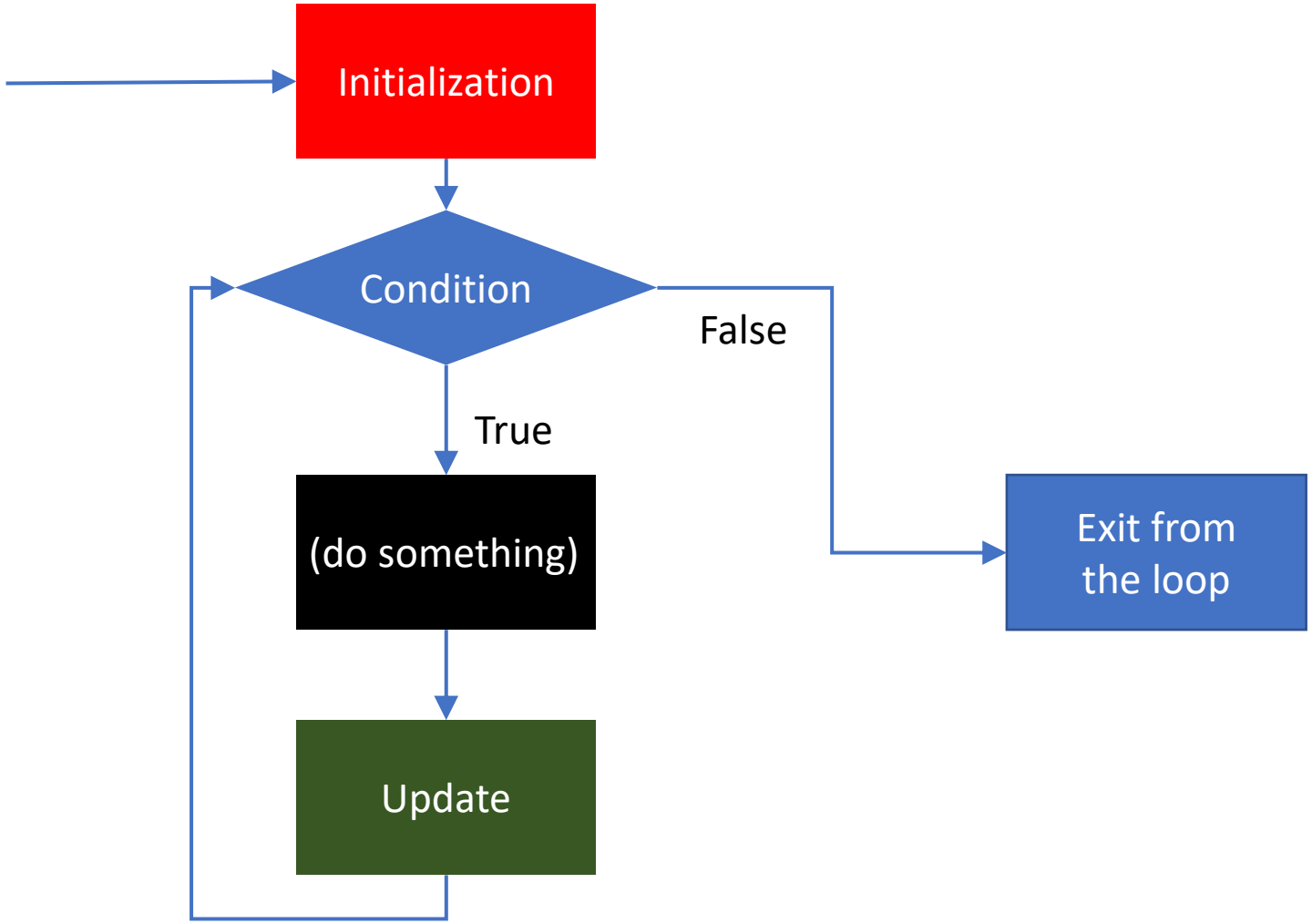
UCLA

Today…

- C++ Loops
  - Exercises: Double Loop, Digits, Grade Calculator ver.2
- Quick HW1 Review
- More Advanced/Challenging Exercise on Loops, *or* HW2 Hints
  - Note: HW2 hints will be given on Thursday anyway

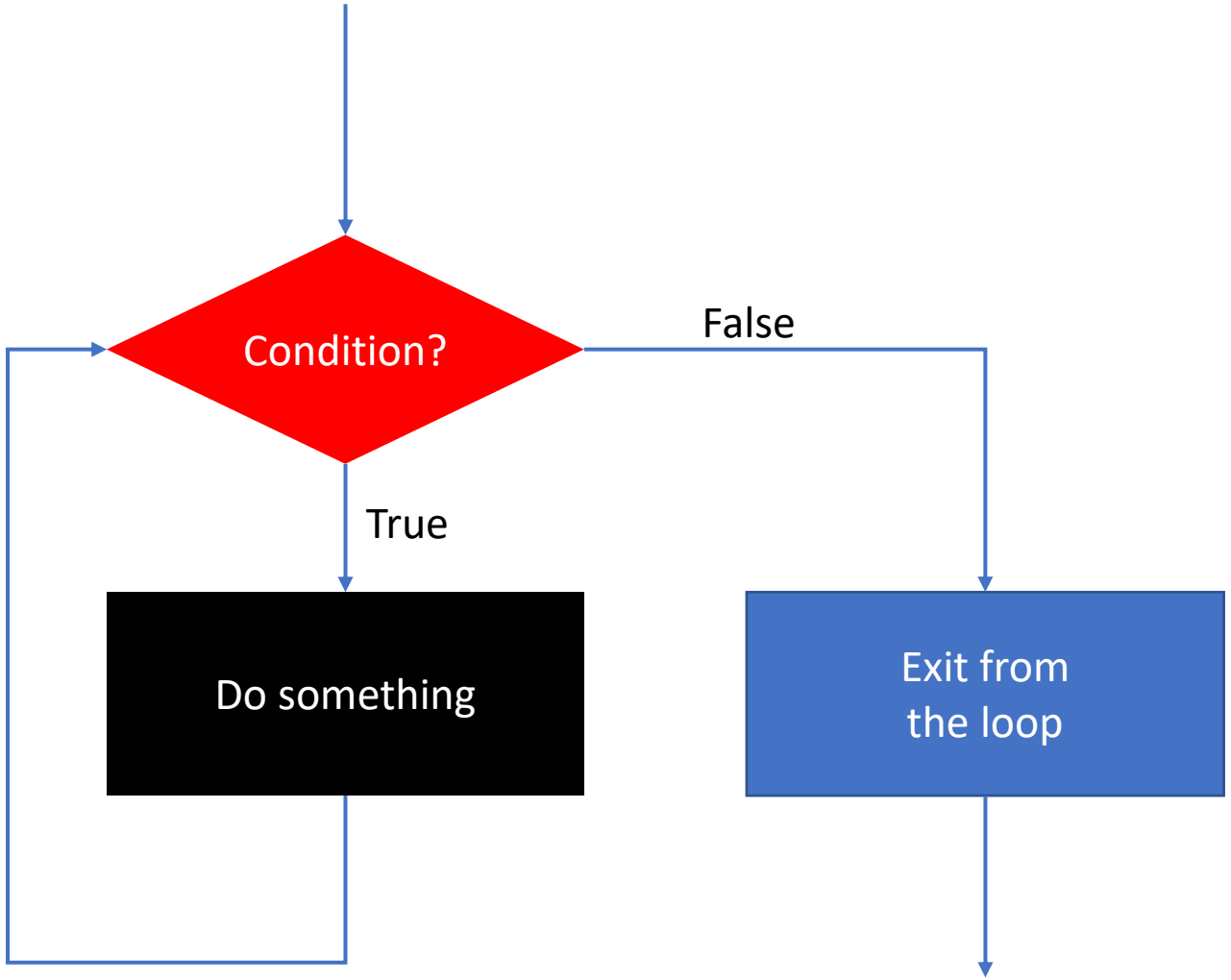
Review on `for`, `while`, and `do while`

- Flow charts



Review on `for`, `while`, and `do while`

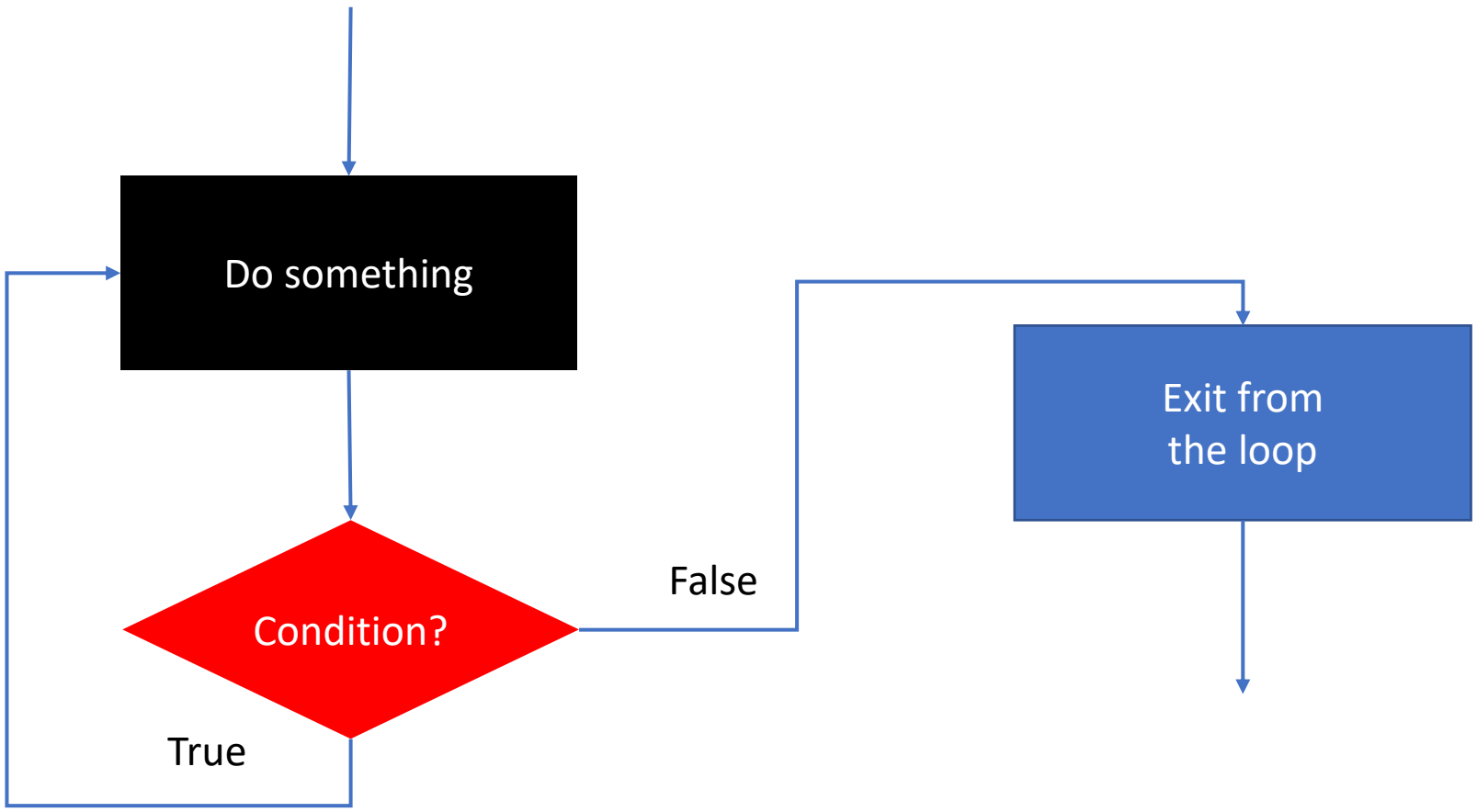
- Flow charts



UCLA

Review on `for`, `while`, and `do while`

- Flow charts



UCLA

Exercise – Double Loop

- Predict the output of the code below:

```
cout << "start" << '\n';
for (int i = 1; i < 4; ++i) {
    for (int j = 1; j < i; ++j) {
        cout << "*";
    }
    cout << '\n';
}
cout << "end";
```

```
start

*
**
end
```

UCLA

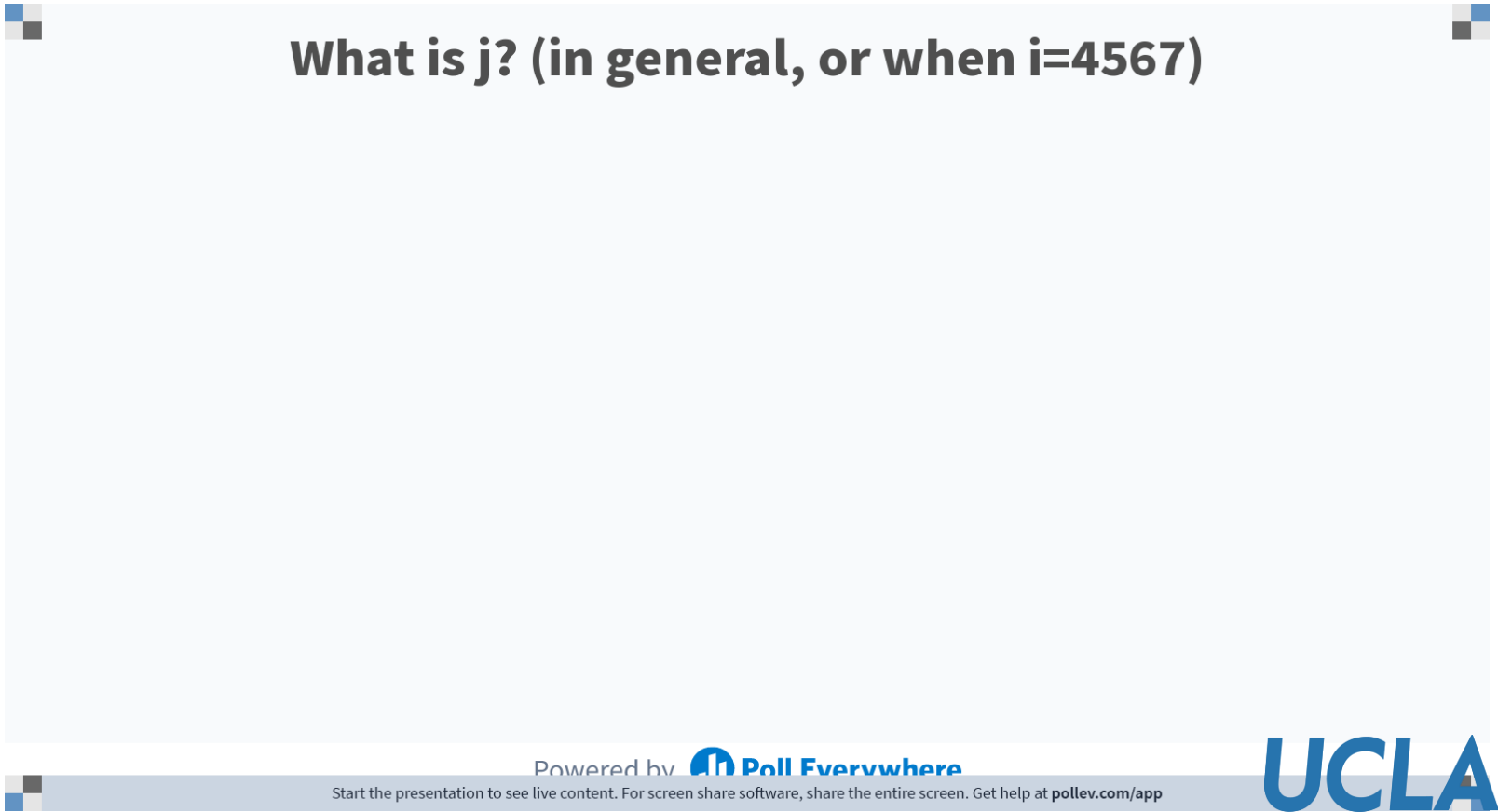
Exercise – Digits

- Describe the value of `j` in terms of `i` after the for loop:

```
int i = (some positive integer);
int j;
for (j = i; (j / 10) > 0; j /= 10);
```

- Hint) When `i = 4567`, what is `j`?

- Ans) `j` represents the first digit of `i`.
  For instance,
  when `i = 4567`, `j = 4`.



Grade Calculator ver. 2 (with Loops)

- Write a program that will calculate a student's final score in some class on the following dual grading system:

User inputs **N**

- Assume that there will be **N** homework assignments total, and the lowest homework score will be dropped.

Scheme A	Scheme B
Midterm Exam 30%	Midterm exam score dropped
Final Exam 40%	Final Exam 70%
Homework 30%	Homework 30%

- The maximum of the two scores obtained from the two schemes will be the final score
- In addition to printing the final score, you should also determine the letter grade based on the following scale: `90 <= A <= 100, 80 <= B < 90, 70 <= C < 80, 60 <= D < 70, 0 <= F < 60.`
- Input and output should be exactly of the following format:

```
Please enter the midterm score (0 - 100): 84.0
Please enter the final exam score (0 - 100): 99.0
Please enter the number of homework assignments (3 - 10): 3
Please enter the homework 0 score (0 - 100): 44.0
Please enter the homework 1 score (0 - 100): 55.0
Please enter the homework 2 score (0 - 100): 66.0
Your final score based on Scheme A is 82.95
Your final score based on Scheme B is 87.45
 Your final score is 87.45
Your course grade is B
```

Q: Can we replace "if - elseif - else" clauses with *one single expression*?
(HINT: consider Boolean expressions as numeric values)

UCLA

Quick HW1 Review

- Exercise 1: `std::sqrt(width * width + height * height);` for the diagonal
- Exercise 2: Num2Month

```
const string str_f =
"January  "
"February "
"March    "
"April    "
"May      "
"June     "
"July     "
"August   "
"September"
"October  "
"November "
"December ";
cout << "Month: " << str_f.substr(9 * (input - 1), 9) << endl;
const int len_each = 9; // avoid "magic numbers." Below is better
cout << "Month: " << str_f.substr(len_each * (input - 1), len_each) << endl;
```

This is a C++ string literal trick for a very long string
(ignores new lines and concatenates all a a single string literal)

UCLA

HW2 Hints

- Exercise 1: Consider three `if` clauses (you don't need `else`. why?)
- Exercise 2: Num2Rome → a bit more complicated
- But as we did in HW1, let's first think about a simplified problem:
  - `num` is between 1 and 9, inclusive, and we want to convert num into a Roman numeral
  - Four cases, basically:
    - `1 <= num < 4`: Print `I` `num` times
    - `num == 4`: Print `IV`.
    - `5 <= num < 9`: Print `V` first, and then `I` appropriate times (how many times?)
    - `num == 9`: Print `IX`.

  - If you can solve this simplified version, now you can extend this method.
  - HINT: Start with thousands digit first. It's the easiest one (`num<4000`)
    - Then go for the hundreds digit, and tens digit, and finally ones digit (natural to add strings *on the right*)
    - If you want to do it *backwards*, consider using the following operator:
    - `output = [something_to_add] + output;` // add a string *on the left*

UCLA

HW2 Hints: a smarter (but more difficult) method

- Challenge for Exercise 2:
- Can you solve this only using 3 if's and 2 while's?
- HINT: Consider the following helper variables:

```
string romans = "IVXLCDM"
int place = 0;
```

- And notice that
  - I, X, C, M → 0th, 2nd, 4th, and 6th, respectively
  - V, L, D → 1st, 3rd, and 5th, respectively

UCLA

Your Feedback is welcome

- Don’t hesitate to give a feedback on the discussion
- You can use a link in BruinLearn (Google Form)

Click this link

At this point, you may want to request
“more hw hints”,
“more challenging examples/exercises”,
“more details on the classroom examples”, etc.
Please let me know what you want!